

KAIRA CLI -- Portfolio Demo Video Guide

Essential Commands, Step-by-Step Recording Flow, and Spoken Script for Maximum Impact

Executive Summary & Goal: Create a concise **2.0 to 2.5-minute** showcase video demonstrating how Kaira transforms tedious FastAPI boilerplate into clean, production-ready 5-layer architecture in seconds. Focus **strictly on the essentials**: initialization, 5-layer pipeline generation, model relationships, zero-config migrations, and instant live Swagger API testing.

1. High-Impact Video Timeline & Structure (2:30 Total)

Timestamp	Demo Stage	Key Action & Command	Visual Focus
0:00 - 0:20	Hook & Problem	FastAPI boilerplate friction -> Introduce Kaira CLI	Terminal title / clean workspace
0:20 - 0:50	1. Init Project	kaira init blog_api --db sqlite --auth jwt	5-layer project structure generated
0:50 - 1:25	2. Generate 5-Layers	kaira generate model User --fields "..." kaira generate model Post --fields "..."	Model, Repo, Schema, Service, Router
1:25 - 1:50	3. Relate & Migrate	kaira add relation Post --has-many Comment kaira migrate make & run	Zero-config Alembic migrations applied
1:50 - 2:20	4. Live Swagger Test	uvicorn main:app --reload -> /docs	Browser side-by-side: Create User / Posts
2:20 - 2:40	5. Pro Diagnostic & Outro	kaira db info / kaira api export -> GitHub outro	Terminal stats + GitHub repo link

2. Step-by-Step Command Playbook (Part 1)

STEP 1: Initialize a Production FastAPI Project with JWT & SQLite

Showcases instant scaffolding of architecture, async database session, JWT auth endpoints, and security middleware.

```
kaira init blog_api --db sqlite --auth jwt
cd blog_api
```

On-Screen Callout: Point out core/database.py, middleware/security.py, and routers/auth.py in VS Code.

STEP 2: Generate Full 5-Layer Pipelines (User & Post)

In two fast commands, create 10 production files covering models, schemas, repositories, services, and routers.

```
kaira generate model User --fields "username:str, email:str, is_active:bool"
kaira generate model Post --fields "title:str, content:str, published:bool"
```

On-Screen Callout: Highlight routers/user_router.py with dependency injection and services/user_service.py business logic.

2. Step-by-Step Command Playbook (Part 2)

STEP 3: Add Relationships & Zero-Config Database Migrations

Demonstrate AST-powered relationship wiring, followed by automatic Alembic migration creation and execution.

```
kaira generate model Comment --fields "content:str, author_name:str"
kaira add relation Post --has-many Comment --cascade "all, delete-orphan"
kaira migrate make "initial schema"
kaira migrate run
```

On-Screen Callout: Emphasize zero manual Alembic setup needed -- tables are created in SQLite instantly.

STEP 4: Run Live Server & Test Interactive Swagger UI

Boot the FastAPI dev server and demonstrate working CRUD endpoints directly in the interactive documentation.

```
uvicorn main:app --reload
# Open browser at http://127.0.0.1:8000/docs
```

On-Screen Callout: Execute POST /api/v1/users/ and GET /api/v1/posts/ to show valid 200 OK responses.

STEP 5: Showcase 1 Pro CLI Feature & Outro

Run a quick diagnostic or export command to demonstrate developer tooling depth before concluding.

```
kaira db info # Displays active database health, tables, and column counts
kaira api export # Exports complete OpenAPI v3 specification to openapi.json
```

On-Screen Callout: Show the clean ASCII table output of database schema health in the terminal.

3. Spoken Voiceover Script (Word-for-Word Guide)

Read this alongside your screen recording or use it as concise voiceover narration (~2 minutes):

[0:00 - 0:20 (Hook)]

"Building backend APIs with FastAPI is great, but writing repetitive boilerplate for models, schemas, repositories, and routes consumes hours. That's why I built **Kaira** -- an automated CLI that scaffolds full production-grade, 5-layer FastAPI backends in seconds."

[0:20 - 0:50 (Init)]

"Let's start by initializing a project: `kaira init blog_api --db sqlite --auth jwt`. In one command, Kaira sets up async database sessions, JWT auth endpoints, SlowAPI rate limiting, Loguru logging, and security middleware in a clean 5-layer layout."

[0:50 - 1:25 (Generate)]

"Now let's create our entities. Running `kaira generate model User and Post` automatically generates typed SQLAlchemy models, Pydantic v2 validation schemas, CRUD repositories, business logic services, and FastAPI routers with dependency injection."

[1:25 - 1:50 (Relate & Migrate)]

"Connecting models is effortless: `kaira add relation Post --has-many Comment` injects the ORM relationship. Then with zero Alembic configuration, `kaira migrate make` and `kaira migrate run` creates and applies the database schema instantly."

[1:50 - 2:20 (Swagger UI)]

"Now we launch with `uvicorn main:app --reload` and head to /docs. Our complete CRUD endpoints and auth flow are live. Let's create a user and fetch posts -- full validation and error handling working out of the box."

[2:20 - 2:40 (Outro)]

"Kaira also features database diagnostics, Redis caching, and OpenAPI client generation. Check out the GitHub repository linked below to learn more. Thanks for watching!"

4. Recording Setup & Portfolio Pro-Tips

- **Screen Layout:** Use a 55/45 split screen -- Dark mode Terminal on the left (55%), VS Code file explorer or Chrome Swagger UI on the right (45%).
- **Terminal Font & Padding:** Increase terminal font size to **16px - 18px** with clean padding so text is crisp even when viewed on mobile screens.
- **Clean State Reset:** Before hitting record, ensure no existing `blog_api` folder exists (run `rm -rf blog_api`) so there are no file overwrite prompts.
- **Typo Prevention:** Keep all commands in a text scratchpad or terminal history so you can execute them fluidly without long typing pauses.
- **On-Screen Overlays:** Add short, bold callout titles in post-editing (e.g. *'5 Layers Generated in 1s'*, *'Zero-Config Migrations'*) for viewers watching on mute.
- **Portfolio Placement:** Export as a high-bitrate 1080p 60fps video, or create a 15-second looping GIF/WebM highlight for the hero header of your portfolio.

5. All-in-One Copy-Paste Command Cheatsheet

Copy and paste these exact commands into your terminal for a smooth dry-run or live record session:

```
# 1. Clean previous run (if any)
rm -rf blog_api

# 2. Initialize project with SQLite & JWT Auth
kaira init blog_api --db sqlite --auth jwt
cd blog_api

# 3. Generate 5-layer pipelines for User, Post, and Comment
kaira generate model User --fields "username:str, email:str, is_active:bool"
kaira generate model Post --fields "title:str, content:str, published:bool"
kaira generate model Comment --fields "content:str, author_name:str"

# 4. Link relationships
kaira add relation Post --has-many Comment --cascade "all, delete-orphan"

# 5. Run zero-config database migrations
kaira migrate make "initial schema with blog entities"
kaira migrate run

# 6. Inspect database structure (Optional CLI flex)
kaira db info

# 7. Launch development server & open docs
uvicorn main:app --reload
# Browser: http://127.0.0.1:8000/docs
```

Portfolio Takeaway: Following this blueprint will result in an engaging, professional developer showcase that demonstrates architectural expertise, clean CLI design, and full-stack backend automation in under 3 minutes.